

YOUNGSTOWN-WARREN, OH, USA

WMO: 725250

Lat: 41.255N

Lon: 80.674W

Elev: 356

StdP: 97.12

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14852

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-13.8	-20.7	0.6	-15.3	-18.3	0.8	-12.8	11.3	-2.7	10.5	-2.3	3.8	240	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	31.2	22.6	29.8	21.8	28.5	20.9	23.9	29.1	23.0	27.6	22.2	26.6	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.8	26.5	21.5	16.9	25.4	20.8	16.1	24.6	73.3	29.2	69.8	27.6	66.7	26.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.4	7.6	-19.5	33.1	4.0	1.6	-22.4	34.2	-24.7	35.2	-27.0	36.1	-29.9	37.3		
(5)			-19.9	25.4	3.9	1.0	-22.7	26.1	-24.9	26.7	-27.1	27.3	-30.0	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-3.2	-1.7	2.8	9.5	15.1	19.7	21.8	21.0	17.4	11.3	5.2	0.0
	DBStd	10.11	6.76	6.41	6.39	5.43	4.78	3.76	3.00	2.96	3.96	4.76	5.24	5.48
	HDD10.0	1582	411	332	241	73	10	0	0	1	42	161	312	
	HDD18.3	3396	667	562	482	268	124	29	6	9	63	225	393	568
	CDD10.0	1572	2	3	17	59	168	292	365	341	224	81	18	3
	CDD18.3	344	0	0	1	4	24	71	112	92	36	6	0	0
	CDH23.3	2851	0	0	5	47	229	595	952	706	282	34	1	0
	CDH26.7	769	0	0	0	6	47	167	291	187	68	4	0	0
(14) Wind	WSAvg	3.7	4.5	4.3	4.3	4.3	3.6	3.2	3.0	2.8	3.1	3.6	4.0	4.2
(15) Precipitation	PrecAvg	981	65	57	78	85	93	98	106	86	91	72	77	76
	PrecMax	1374	149	148	162	185	212	271	264	197	201	155	231	166
	PrecMin	604	18	14	27	26	20	18	17	13	7	11	21	25
	PrecStd	153	32	28	31	36	41	49	51	42	43	32	36	29
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	18.2	23.4	27.6	30.0	32.1	33.0	32.4	31.0	26.7	20.8	16.7
		MCWB	12.8	12.6	15.3	17.2	20.7	22.9	23.7	23.5	22.1	19.8	14.4	13.2
	2%	DB	13.3	13.8	19.4	24.5	28.0	30.2	31.2	30.4	28.7	23.7	18.1	13.6
		MCWB	11.4	10.2	12.7	15.4	19.3	21.9	22.8	22.2	20.9	17.6	13.2	11.0
	5%	DB	10.0	10.5	16.2	21.8	26.1	28.7	29.7	28.7	26.5	21.6	16.0	11.0
		MCWB	8.1	7.3	11.1	14.1	18.5	21.1	21.9	21.4	19.9	16.5	11.8	8.8
	10%	DB	6.5	7.3	13.2	19.0	23.9	26.9	28.2	27.2	24.6	19.2	13.7	8.1
		MCWB	4.6	4.5	9.2	12.4	17.0	20.0	21.0	20.5	18.8	15.1	10.3	6.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	13.9	15.9	18.5	22.2	24.1	25.4	24.9	23.9	21.4	16.4	14.2
		MCDB	15.4	17.0	21.9	24.7	28.2	30.0	30.8	30.3	28.8	25.2	19.3	16.0
	2%	WB	11.6	10.7	13.9	16.7	20.7	23.0	24.1	23.5	22.3	19.3	14.1	11.5
		MCDB	13.2	13.1	17.9	22.5	25.9	28.5	29.3	28.0	26.7	22.2	16.5	13.1
	5%	WB	8.1	7.5	11.8	15.2	19.5	22.1	23.1	22.6	21.1	17.5	12.5	8.8
		MCDB	9.7	10.1	15.4	20.1	24.5	27.1	27.7	26.5	24.8	20.3	15.4	10.3
	10%	WB	4.6	4.8	9.2	13.5	18.3	21.1	22.2	21.8	19.9	15.6	10.5	6.1
		MCDB	6.3	7.1	13.0	18.0	22.7	25.3	26.3	25.6	23.1	18.6	13.4	8.1
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.4	9.5	11.5	11.6	11.1	11.2	11.0	11.1	9.9	8.2	6.7
		MCDBR	10.2	12.1	13.4	14.9	13.7	12.6	12.3	12.4	12.9	12.6	11.2	10.0
	5% WB	MCWBR	8.8	9.6	8.5	8.6	6.9	6.1	5.5	5.6	6.1	7.1	7.7	8.6
		MCWBR	9.7	11.3	11.9	12.7	11.8	11.0	10.6	10.3	10.6	10.6	10.1	9.6
(40) Clear-Sky Solar Irradiance	taub		0.299	0.329	0.353	0.383	0.421	0.443	0.448	0.435	0.383	0.359	0.316	0.298
		taud	2.437	2.334	2.345	2.305	2.214	2.189	2.194	2.222	2.346	2.411	2.511	2.489
	Ebn at Noon		857	875	894	888	857	836	828	827	851	832	832	829
		Edh at Noon	78	101	112	125	141	144	142	135	111	93	73	69
(44) All-Sky Solar Radiation	RadAvg		1.40	2.29	3.25	4.43	5.27	5.75	5.80	5.18	4.17	2.69	1.75	1.13
	RadStd		0.13	0.27	0.33	0.45	0.47	0.48	0.43	0.28	0.41	0.26	0.25	0.11

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (42 neighbors)										
		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+55
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+55

Nomenclature: See separate page